Welcome to

instats

The session will begin shortly.

Start

SECOND MOVEMENT, 35 MINUTES

The validity check.

Thirty-row gold set. Confusion matrix. Cohen's kappa. Pearson r. Subgroup table.

02

A COMMON QUESTION

Why thirty paragraphs.

TOO FEW

Five

Cannot distinguish a sixty-percent accurate model from an eighty-percent accurate one. The standard error on accuracy at n=5 is enormous.

SWEET SPOT

Thirty

Big enough to compute a usable kappa with a reasonable confidence interval. Small enough that two coders can finish in an afternoon. Stratify by year and you can spot subgroup drift early.

TOO MANY FOR FIRST CUT

Three hundred

A week of careful work. You will not actually do it before stopping to look at results. Better to start with thirty and add another thirty if the first round leaves you uncertain.

The thirty-paragraph gold set is the iteration unit. Code, look, refine, code again.

STEP ONE. WRITE THE CODEBOOK.

Three rules for our climate-stance task.

RULE 01, +1

Praises stronger international climate action

No offsetting skepticism. Examples. Endorsing Paris. Praising COP outcomes. Calling for higher emissions targets.

RULE 02, -1

Skeptical of stronger international climate action

No offsetting praise. Examples. Rejecting binding targets. Defending fossil-fuel investment as a national priority.

RULE 03, 0

Other

Mixed. Off-topic. Aspirational without evaluation. Most of the corpus lands here.

Three rules. Operational. Disjoint. Testable. Print this on a sticky note.

Five paragraphs through the codebook.

PARAGRAPH (PARAPHRASED)	SCORE	WHY
Reaffirmation of Paris with a call for more ambitious nationally determined contributions.	+1	Pure praise.
Rejection of binding targets imposed on developing economies, framed as climate justice.	0	Mixed. Document the call.
List of ratifications of the Glasgow Climate Pact.	+1	Pure praise.
Description of national fossil-fuel infrastructure as priority for energy security.	-1	No offsetting climate language.
Speech entirely about Security Council reform.	0	Off-topic.

Pre-coded gold labels for the demo are bundled in `demo.ipynb` Section D. In a real project, you replace this column with your own hand codes.

Confusion matrix and Cohen's kappa.

WHAT YOU SEE LIVE

The confusion matrix compares the off-the-shelf zero-shot prediction to the hand-coded gold label on the thirty-row sample.

Diagonal cells are agreement. Off-diagonal cells are disagreement. Diagonal totals between two-thirds and three-quarters are typical for off-the-shelf zero-shot on a three-way climate task.

Cohen's kappa adjusts accuracy for chance agreement. Read whatever the live number is and treat it as the headline.

KAPPA THRESHOLDS, LANDIS AND KOCH 1977

KAPPA	VERDICT
> 0.80	Almost perfect
0.60 to 0.80	Substantial
0.40 to 0.60	Moderate
0.20 to 0.40	Fair
< 0.20	Poor

Off-the-shelf on this corpus typically lands 0.40 to 0.70. The single best one-number summary of model-versus-human agreement.

Pearson r and Spearman rho use the continuous signal.

WHY BOTH KAPPA AND R

Different failures. Kappa low and r high means the argmax cutoff is wrong but the model knows what it is looking at. Kappa high and r low means the argmax is correct but probabilities are noisy.

- **Pearson r.** Continuous supportive probability versus hand-coded score. Can be 0.5 to 0.7 for off-the-shelf zero-shot.
- **Spearman rho.** Rank correlation. Robust to outliers and non-linear monotone shifts.
- **Report both.** They typically agree to within a couple of points.

```python
from scipy.stats import pearsonr, spearmanr

r, _   = pearsonr(p_supportive, gold_score)
rho, _ = spearmanr(p_supportive, gold_score)

print(f"Pearson r    {r:+.2f}")
print(f"Spearman rho {rho:+.2f}")
```

Both numbers belong in your appendix. Headline kappa for the binary call. Pearson r for the continuous signal underneath.

One last validity check. Subgroup performance.

SPLIT	SUBGROUP	N	SHARE SUPPORTIVE	MEAN P(SUPPORTIVE)
Era	pre-2015	~3000	illustrative	illustrative
	post-2015	~2000	illustrative	illustrative
Region	Europe	~700	illustrative	illustrative
	Africa	~1500	illustrative	illustrative
	Oceania	~250	illustrative	illustrative
	Americas	~900	illustrative	illustrative
	Asia	~1300	illustrative	illustrative

If kappa varies by more than 0.15 across subgroups, you have a comparability problem.

Numbers in the table are illustrative placeholders. Live values fill the slide once the demo cell finishes.

STILL MOVEMENT 02, WHEN THE KAPPA IS TOO LOW

Beat the ceiling with the labels you already have.

No new data. The thirty paragraphs you just hand-coded become training data. Fine-tune in two minutes with SetFit.

Zero-shot asks. Few-shot learns.

ZERO-SHOT, WHAT WE DID ALL MORNING

A generic NLI model judges whether your label sentence is entailed by the text. No training. Fast to stand up. The wording of the label is the whole model.

FEW-SHOT FINE-TUNING

Teach the model your specific task from a handful of examples. The label wording stops mattering. The examples do the work.

WHY SETFIT OVER CLASSIC FINE-TUNING

Learns from eight examples per class

Classic fine-tuning wants hundreds of labels. SetFit fine-tunes a sentence-embedding model with contrastive pairs, so it learns from as few as eight examples per class.

Thirty hand-codes is enough to try it. That is exactly what the gold set gives you.

Architecturally the two differ. The DeBERTa-NLI model from this morning is a cross-encoder. It reads your text and your candidate label together and judges entailment, and it never learns your task. SetFit sits on a sentence-transformer, a bi-encoder that maps each text to a single embedding vector. Fine-tuning pulls same-label vectors closer, then a small head reads them. The examples do the work the label wording did before.

THE MOVE

The gold set was never just for validation.

SPLIT THE THIRTY

— **~20 to train.** Two thirds of the gold set, stratified by label.

— **~10 held out.** One third, never seen in training, scored honestly.

— **Same labels, two jobs.** Validation this morning. Training data this afternoon.

ONE HONEST WARNING

Thirty labels is a proof of concept

Enough to demonstrate the method, often enough to beat zero-shot. Not enough for a final published number. The held-out set is tiny, so the kappa you read is noisy.

Collect a hundred or more labels before you trust the gain.

Five lines to fine-tune.

```
from setfit import SetFitModel, Trainer
from datasets import Dataset

train, test = train_test_split(gold,
  test_size=0.34, stratify=gold.gold_label,
  random_state=7)

ds = Dataset.from_pandas(train.rename(
  columns={"gold_label": "label"}))

ft = SetFitModel.from_pretrained(
  "sentence-transformers/"
  "paraphrase-MiniLM-L6-v2")

Trainer(model=ft, train_dataset=ds).train()
```

WHAT IT IS DOING

SetFit nudges the embedding space so paragraphs with the same label sit closer together, then fits a light classifier head on top.

No giant gradient update to a 400-million-parameter model. That is why it runs on a free GPU.

About two minutes on a T4.

Did it beat the ceiling?

SCORED ON THE SAME HELD-OUT GOLD ROWS	COHEN'S KAPPA	ACCURACY
Off-the-shelf zero-shot, the number from before the break	**0.52**	0.70
SetFit, fine-tuned on ~20 of your own labels	**0.71**	0.80

Numbers are illustrative. Live values fill the slide once the comparison cell finishes. On thirty labels SetFit usually edges out zero-shot, sometimes by a wide margin, sometimes within the noise.

The takeaway is the workflow, not the single number. Hand-code, validate, and fine-tune with the labels you already have.

THIRD MOVEMENT, 35 MINUTES

Climate emphasis over time.

Country-year scores. Time series 1946 to 2025. Compare against the Paris Agreement.

03

A PUBLISHED-STYLE MEASURE

One small piece of a much-built family.

The plot we are about to build is the kind of measure that has been constructed from UNGDC many times. Baturo, Dasandi, and Mikhaylov 2017 introduced UNGDC and demonstrated several text-as-data measures on it. More recent papers have extracted topic emphasis, sentiment toward institutions, and stance toward specific issues from the same corpus.

What we are doing today is one small piece of that family. Climate-emphasis share by country and year, computed from a zero-shot encoder, plotted from 1946 to 2025, with the Paris Agreement marked.

WHAT WE ARE NOT DOING

— Causal claims about Paris Agreement effects.

— Event-study estimation.

— A complete replication of any single published proxy.

The plot is a face-validity check, not an event study. The methodology is the point.

THREE WAYS TO AGGREGATE

None is neutral.

OPTION 01

Mean P(supportive)

Average the supportive-class probability across all speeches in a country-year. Sensitive. Uses the continuous signal.

OPTION 02

Share supportive

Count the share of speeches where the argmax is supportive. Simpler. Throws away the underlying probability.

OPTION 03

Top-K

Keep only the most extreme K speeches per country-year. Useful when most speeches are administrative. K is a hyper-parameter. Validate it against the gold set before publishing.

For "is climate language increasing on average," mean works. For "are some countries pivoting hard in particular years," top-K is more honest. Document the choice in the appendix and run a robustness check with one alternative.

Two views of the time series.

VIEW 01, GLOBAL MEAN

Mean of supportive-class probability across all speeches, by year, 1946 to 2025.

Low through the 1970s and 1980s. Ticks up through the 1990s. Accelerates in the early 2000s. Visible bend around 2015.

Vertical line marks the Paris Agreement signing in December 2015.

VIEW 02, REGIONAL SPLIT

Same curve broken out by UN region. Africa, Americas, Asia, Europe, Oceania.

Oceania often leads. Small island states have been making climate central to UN speeches since the late 1980s. Europe joins them visibly post-Paris. Asia and the Americas track the global mean. Africa is more volatile.

The Oceania line is the moment when off-the-shelf, with thirty seconds of label engineering, produces a substantively informative time series.

The slope around Paris is what a reviewer wants to see.

WHAT THE CURVE DOES

- Pre-Paris years (2010 to 2014). Roughly flat trend.
- 2015 to 2017 stretch. Visible bend upward.
- Post-2017. Curve flattens at a higher level.

The Paris Agreement was signed in December 2015 and entered into force in November 2016. The 2016 General Assembly is the first one fully after Paris.

WHAT THIS IS AND IS NOT

Is. A face-validity check. The curve is doing what a sensible measure should do near a major external event in the same domain.

Is not. An event-study estimate. We have not adjusted for country composition, speech length, or shift in topic prevalence outside climate. Doing those adjustments is the next step in a real paper.

If the picture is wrong, fix the measure before estimating any effect.

Two columns.
Must-have. Often-asked-for.

MUST-HAVE

- Codebook reproduced in full, dated.
- Gold-set sample size and selection method.
- Cohen's kappa, accuracy, Pearson r, Spearman rho versus gold.
- Confusion matrix.
- Model ID with version pin. Candidate-label list. Aggregation rule.

OFTEN ASKED FOR IN REVISIONS

- Subgroup performance by era and region.
- Sensitivity to candidate-label wording.
- Comparison against a simpler baseline (keyword search, lexicon).
- Notebook and gold-set CSV in replication package.

If the appendix has all of column one, the proxy is defensible.

Two rungs beyond the SetFit demo.

RUNG 01, NO EXTRA TRAINING DATA

Hypothesis ensembling

Replace one supportive label with three or five differently worded versions of the same hypothesis. Average their probabilities. The wording-sensitivity collapses.

Kappa typically rises by 0.05 to 0.10. No new training data. Days of work, not weeks.

RUNG 02, THREE HUNDRED HAND-CODES

Full supervised fine-tuning

When SetFit on a handful of labels is not enough, hand-code 300 paragraphs and fine-tune a DeBERTa-v3-base or v3-large head on them.

Kappa typically rises by 0.10 to 0.20. The cost is a week of hand-coding. The benefit is a measure that can survive serious validation scrutiny.

The ladder is the point. Off-the-shelf zero-shot first. SetFit on your validation labels when the ceiling bites. These two rungs when the stakes justify the labels.

Open chat.
Nine minutes.

PRE-ANSWERED	SHORT ANSWER
Can I run today's pipeline on free Colab CPU?	Yes, but the 4,761-row pass takes about 30 minutes instead of three. Use a smaller sample for the first run.
What if my corpus is in another language?	Translate first, or pick a multilingual encoder like XLM-RoBERTa-NLI. Both have tradeoffs.
How big can the encoder be on free GPU?	DeBERTa-v3-base, the model we used, fits free T4 with room. v3-large also fits, but slower. Above 1B parameters needs paid Pro.
Where do I publish my replication notebook?	Code on OSF or GitHub. Data on Harvard Dataverse or your institutional repo. Ship the gold-set CSV with the code.

THE VALIDATION CHECKLIST AS ONE SLIDE

Five lines.
Memorize them.

— One codebook with three rules.

— Thirty hand-coded paragraphs.

— One confusion matrix.

— Cohen's kappa, accuracy, Pearson r, Spearman rho.

— One subgroup table.

If you have those five things in the appendix, your encoder-based measure is defensible.

TOMORROW

Three jobs encoders cannot easily do.

JOB 01

Free-text extraction

Pull structured fields from a press release, even when the fields are not labels you can list in advance.

JOB 02

Normalization

Resolve "USA," "United States," and "U.S." to a single ISO code across messy crawled text. Encoders are bad at this. Generative models are good at it.

JOB 03

Lightweight crawling

Pull paragraphs off a public site, store the raw HTML for reproducibility, feed the cleaned text to a generative model. End-to-end pipeline.

Same reproducibility demands as today. New tools.

END OF DAY 2

You turned a notebook into a measurement.

A codebook with three rules. A thirty-paragraph hand-coded gold set. A confusion matrix. A country-year CSV plotted against the Paris Agreement. A validation appendix that satisfies most political-science reviewers.

OPTIONAL READING BEFORE DAY 3

Adcock and Collier (2001, APSR), "Measurement Validity, A Shared Standard for Qualitative and Quantitative Research." The framework decides which of today's checks belong in the paper and which stay in the notebook.

Stop